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WINDING MACHINE WITH AUTOMATIC INTRODUCTION OF REINFORCING SHEET

Abstract

A device is for automatically introducing a reinforcing sheet into a coreless coil applicable to a coreless winding machine. The winding machine includes such a device. In particular, a device is for introducing a reinforcing paper sheet into a coreless roll being wound with a continuous strip material by a coreless winding machine. The device includes a feeding tray for paper sheets, a feeder of the paper sheets taken from the feeding tray, and an introduction member of a paper sheet loaded into a roll being wound in the winding machine.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATION

[0001] This application claims priority to and benefit of Italian Patent Application No. 102024000005356 filed Mar. 11, 2024, the contents of which are incorporated by reference in their entirety.

TECHNICAL FIELD OF THE INVENTION

[0002] The present invention relates to a device for automatically introducing a reinforcing sheet into a coreless coil applicable to a winding machine of the coreless type, and to the winding machine comprising such a device.

PRIOR ART

[0003] The prior art of coils for household use, e.g., coils of aluminum foil or paper, involves winding the continuous strip material over a cardboard core, the function of which is to act as a support for the wound material.

[0004] Such an operation is conducted by appropriate winding machines, which produce the rolls of aluminum, paper (e.g., baking paper) or other materials from a feeding coil of considerable size, supporting the cardboard core downstream on a roller or appropriate grippers rotating at a winding speed which can even be very high.

[0005] However, the use of the cardboard core has several disadvantages; firstly, the cost, but also the fact that the cardboard cores must be manufactured or purchased from third parties and thus must be stored, requiring, due to their very shape and fragility, space and particular care for storage.

[0006] In order to solve such a problem, winding machines of the coreless type have been manufactured, i.e., capable of winding the continuous strip material directly about a machine mandrel, from which the formed rolls can then be easily pulled out. An example of such machines that has proven to be particularly efficient is that described in EP 3810536 B1 to the same Applicant.

[0007] This solution, although satisfactory in countless respects, has exhibited a criticality in some cases, e.g., according to the type and length of wound material, resulting in rolls of insufficient crushing strength.

[0008] Thus, there is a need to obtain rolls of continuous strip material, in particular aluminum or paper, which have sufficient crushing strength but do not require the use of a cardboard core.

SUMMARY OF THE INVENTION

[0009] The problem outlined above is solved by a device for automatically introducing a reinforcing sheet and a winding machine which comprises such a device, as defined in the appended claims, the definitions of which form an integral part of the present description.

[0010] The invention also relates to a method for automatically introducing a reinforcing sheet into a roll of a continuous strip material, and the reinforced roll thus obtained.

[0011] In particular, the present invention relates to: [0012] 1) a device for introducing a reinforcing paper sheet into a coreless roll being wound with a continuous strip material by means of a winding machine of the coreless type, comprising: [0013] a feeding tray for paper sheets, [0014] a feeder of the paper sheets taken from the feeding tray, [0015] an introduction member of a paper sheet loaded into a roll being wound in the winding machine; [0016] 2) a device according to item 1, wherein the introduction member of the sheet comprises a first conveyor member and a second conveyor member placed in sequence along an inclined path, wherein the first conveyor member is inclined with a gradient less than the second conveyor member and wherein, preferably, the angle formed by the plane of the second conveyor member with respect to a horizontal plane is greater than 45°, more preferably greater than or equal to 70°; [0017] 3) a device according to item 2, wherein both the first and second conveyor members comprise conveyor belts wound about pulleys supported on shafts rotating by independent drives, wherein the drive of the second conveyor member is configured to operate at a speed such as to move the sheet homokinetically with respect to the winding speed of the roll; [0018] 4) a device according to item 2 or 3, wherein the first conveyor member comprises a series of idle rollers arranged at one end of respective arms hinged onto support bars to form gravity pressing means for the paper sheets; [0019] 5) a device according to any one of items 2 to 4, wherein the second conveyor member comprises a belt presser placed above the conveyor belts and comprising belts wound about pulleys supported idly on shafts; [0020] 6) a device according to any one of items 2 to 5, wherein the second conveyor member comprises a proximal end with respect to the first conveyor member and a distal end, wherein the distal end is associated with a guide member for the sheet arranged in a substantially vertical position and comprising a shaft which supports idle pulleys, drive belts being wound on the pulleys and on corresponding pulleys supported idly on a shaft arranged at the distal end of the second conveyor member; [0021] 7) a device according to any one of items 1 to 6, wherein the feeding tray comprises containment edges and an open bottom placed above a proximal end of the feeder, wherein the containment edge placed straddling the feeder ends close to a gripping surface of the feeder without coming into contact therewith, so as to allow the paper sheet to be singularized and taken from a stack of sheets from below; [0022] 8) a device according to any one of items 1 to 7, wherein the feeder comprises an ascending belt system running about motorized pulleys and idle pulleys, wherein the belt system changes the gradient thereof close to a distal end of the feeder, forming a substantially horizontal segment; [0023] 9) a device according to item 8, wherein a plurality of idle rollers is placed above the belt system, at one end of respective arms hinged onto support bars to form gravity pressing means for the paper sheets transiting on the feeder; [0024] 10) a coreless-type winding machine comprising a device according to any one of items 1 to 9, wherein the winding machine comprises: [0025] a coil-holder roller, which rotationally supports a coil of continuous strip material, [0026] a first motorized drive roller and a second motorized drive roller, where the second drive roller is coupled to a pressing roller so that the continuous strip material passes between the two rollers, [0027] a plurality of motorized winding mandrels, placed on a revolver member, the revolver member being discontinuously rotatable, so as to present a different winding mandrel from time to time at the second motorized drive roller, [0028] where the guide member of the second conveyor member of the device ends above the contact point between the second drive roller and the winding mandrel of the roll being wound, [0029] and wherein the winding mandrels have, along the cylindrical surface thereof, fixed circular arc sectors alternating with retractable circular arc sectors, which form a cylindrical surface during the step of winding to form the roll and retract radially during the step of extracting the roll; [0030] 11) a method of winding a coreless roll of a continuous strip material comprising the steps of: [0031] providing a winding machine for coreless rolls of the continuous strip material, [0032] winding on a mandrel of the winding machine a first set of turns of the continuous strip material, [0033] introducing a reinforcing paper sheet into the winding roll while winding a second set of turns, [0034] singularizing the roll thus formed; [0035] 12) a method according to item 10, comprising the step of providing a winding machine of the coreless type comprising a device for introducing a paper sheet as defined in item 10; [0036] 13) a coreless roll of a continuous strip material, preferably food grade aluminum or alloys thereof or food grade paper, consisting of a central hole surrounded by a first set of turns of the continuous strip material and a second set of turns of the continuous strip material, wherein the first set of turns comprises a smaller number of turns than the second set of turns and preferably between 1 and 9 turns, and wherein a paper sheet is placed between the first set of turns and the second set of turns; [0037] 14 a roll according to item 13, wherein the wound paper sheet protrudes from two axial ends of the roll, preferably by 2-7 mm; [0038] 15) a roll according to item 13 or 14, wherein the paper sheet has information, such as advertising lettering, recipes, contest indications, or the like.

Description

BRIEF DESCRIPTION OF THE FIGURES
[0039] FIG. **1** depicts a perspective view of a winding machine comprising the device for automatically introducing a reinforcing sheet according to the invention;
[0040] FIG. **2** depicts a perspective view of the winding machine in FIG. **1** from a different point of view;
[0041] FIG. **3** depicts a top plan view of the winding machine in FIG. **1**;
[0042] FIG. **4** depicts a longitudinal section side view of the winding machine in FIG. **1**;
[0043] FIG. **4***a* depicts an enlarged perspective view of a detail in FIG. **4**;
[0044] FIG. **5** depicts a rear view of the winding machine of the invention taken along direction A in FIG. **3**;
[0045] FIG. **6** is a perspective view of a detail of the device for automatically introducing a reinforcing sheet according to the invention;
[0046] FIG. **7** depicts a perspective view of the detail in FIG. **6** from a different point of view;
[0047] FIG. **8** depicts a front view of an axial end of a roll of continuous strip material obtained with the machine of the invention;
[0048] FIG. **9** depicts a perspective view of an axial end of the roll in FIG. **8**.
DETAILED DESCRIPTION OF THE INVENTION
[0049] With reference to the figures, the winding machine, indicated by reference numeral **1** as a whole, comprises a coreless-type rewinding unit **2**, to which a device **3** for introducing a reinforcing sheet is applied.
[0050] The rewinding unit **1** shown in FIGS. **1-5** is to be considered as a preferred embodiment only. It should be understood that any coreless rewinding unit according to the prior art can equally be used in combination with the device **3** according to the invention. For example, other embodiments of coreless winding machines are described in U.S. Pat. No. 2,394,503, JP 2004161478A or EP 2660173 A1.
[0051] The rewinding unit **1** shown in the figures will be described briefly below because it is not a subject of the present invention per se. For a more detailed description, reference should be made to patent EP 3810536 B1 to the same Applicant.
[0052] Such a rewinding unit **1** is housed in a containment structure **4** which allows separating the external environment from the moving parts of the machine, both for reasons of operator safety and to minimize noise. The containment structure preferably comprises transparent panels to allow the visual inspection of the machine during its operation. External structures **5***a*, **5***b* house the drives and the control panel.
[0053] The rewinding unit **1** comprises a coil-holder roller **6** which rotationally supports a coil of continuous strip material (not shown). The coil-holder roller **6** can be motorized or simply braked to prevent uncontrolled unwinding thereof.
[0054] Typically, such a continuous strip material will be aluminum or alloys thereof, e.g., food-grade aluminum, or paper. However, it cannot be ruled out that such a material also consists of a plastic film or composite material or any other continuous strip material which can be marketed in rolls.
[0055] Two deflection rollers **7***a*, **7***b* are arranged downstream of the motorized coil-holder roller **6** for the continuous strip material S to be rewound. Thus, the continuous strip S runs on a first drive roller **8** and then on a second drive roller **9**, both motorized directly or by means of a transmission, e.g., a belt transmission from the first drive roller **8** to the second drive roller **9**, or vice versa. The second drive roller **9** is coupled to a pressing roller **10**, so that the continuous strip material S passes between the two rollers **9**, **10**.
[0056] The second drive roller **9** is rotatably placed at a motorized winding mandrel **11**, which is

placed on a revolver member **12**. The revolver member **12** rotatably supports, along a peripheral circumference, a plurality of motorized winding mandrels **11** and is, in turn, rotatable discontinuously, so as to display a different winding mandrel **11** at the second drive roller **9** each time. Substantially, when a roll of continuous strip material S has been formed on a first winding mandrel **11**, the revolver member rotates, bringing a new winding mandrel **11** to the second drive roller **9**, while the first winding mandrel **11** with the roll passes in a cutting station **13**, where the roll is singularized.

[0057] As described in greater detail in EP 3810536 B1, the winding mandrel **11** has, along the cylindrical surface thereof, fixed arc sectors alternating with retractable arc sectors, e.g., three fixed and three retractable sectors, which form a cylindrical surface during the step of winding to form the roll. When the roll is to be pulled out of the winding mandrel **11**, downstream of the cutting station the retractable sectors are brought, by means of a cam, to a radially retracted position. This thus results in a loosening of pressure on the inner surface of the coreless roll, which promotes the extraction thereof.

[0058] The second drive roller **9** and the pressing roller **10** are constrained to two oscillating shoulders which, by rotating about the axis of the first drive roller **8**, allow the rollers **9**, **10** to adjust the position thereof to the diameter of the roll being rewound.

[0059] What has been briefly described so far is the known winding machine, in particular the machine described in EP 3810536 B1 (to which full reference is made for a better understanding of the structure and operation thereof), for the production of rolls of continuous material strip of the coreless type.

[0060] However, it has been seen that in some cases it can be convenient to impart reinforcement to the roll thus produced, but without having to revert to the use of a rigid cardboard core as in the traditional art.

[0061] The inventors of the present patent application have thus reached the solution which includes introducing a paper sheet, e.g., an A4, A3 or similar size sheet (according to the size of the roll to be produced), during the formation of the coreless roll and precisely after a number of turns of the continuous strip material S have already been wound. The introduction of the paper sheet will preferably occur after 0.1-0.8 meters of material has been wound onto the winding mandrel **11**. FIGS. **8** and **9** show the roll **50** as formed comprising the paper sheet **51**.

[0062] The paper sheet **51**, by winding itself about the mandrel **11**, will thus form a paper cylinder of sufficient rigidity to strengthen the roll **50**.

[0063] More specifically, the coreless roll **50** of a continuous strip material S consists of a central hole **52** surrounded by a first set of turns **53** and a second set of turns **54** of the continuous strip material S, where the first set of turns **53** comprises a smaller number of turns than the second set of turns **54** and preferably between 1 and 9 turns, and where a paper sheet **51** is placed between the first set of turns **53** and the second set of turns **54**.

[0064] Preferably, as shown in FIG. **9**, the wound paper sheet **51** protrudes from two axial ends **50***a* of the roll **50**, preferably by 2-7 mm. It thus works as a spacer thus acting as a side protection for the roll **50**.

[0065] In some embodiments, the paper sheet **51** can be printed and exhibit advertising inscriptions, recipes, directions for contests or the like.

[0066] The device **3** for introducing the paper sheet **51** into the roll **50** being formed will now be described with reference to FIGS. **1-7**.

[0067] The device **3** for introducing a paper sheet **51** into the roll **50** being wound comprises:

[0068] a feeding tray **31** for paper sheets **51**, [0069] a feeder **32** of the paper sheets **51** taken from the feeding tray **31**, [0070] an introduction member **33** of the paper sheet **51** into the roll **50** being formed.

[0071] The feeding tray **31** and the feeder **32** are of known type.

[0072] The feeding tray **31** comprises containment sides **34** and an open bottom **35** placed above a proximal end **32***a* of the feeder **32**. The containment side **34** placed straddling the feeder **32** ends close to a gripping surface of the feeder **32** without coming into contact therewith, so as to allow the paper sheet to be singularized and picked from underneath from a stack of sheets present in the feeding tray **31**. Such a mechanism is quite similar to that present, for example, in printers or copiers for household or professional use.

[0073] The feeder **32** comprises an ascending belt system **36** which runs about motorized pulleys **37** and idle pulleys **38**. Close to the distal end **32***b* of the feeder **32**, the belt system **36** changes gradient by means of the aid of appropriate deflection rollers **39** (FIG. **4**), forming a substantially horizontal segment **32***c*.

[0074] A plurality of idle rollers **40** is placed above the belt system **36**, at one end of respective arms **41** hinged onto support bars **42** to form (gravity) pressing means for the paper sheets transiting on the feeder **32**.

[0075] The feeder **32**, at the distal end **32***b*, is connected to the introduction member **33** of the sheet **51**, e.g., by means of braces **43** (FIGS. **6** and **7**).

[0076] The introduction member **33** of the sheet **51** comprises a first conveyor member **44** and a second conveyor member **45** placed sequentially along an inclined path, where the first conveyor member **44** is inclined with a smaller gradient than the second conveyor member **45**. In other words, the angle formed by the plane of the first conveyor member **44** with respect to a horizontal plane is smaller than the angle formed by the plane of the second conveyor member **45** with respect to the same horizontal plane, the latter being greater than 45°, preferably greater than or equal to 70°.

[0077] Both the first and second conveyor member **44**, **45** comprise conveyor belts **46** wound around pulleys **47** supported on shafts **48** rotating by independent drives **49**. As will be better described below, the drives **49** are configured to operate discontinuously, and the drive **49** of the second conveyor member **45** is configured to operate at a speed such that it moves the sheet **51** homokinetically with respect to the winding speed of the roll **50** on the winding mandrel **11**.

[0078] Like the feeder **32**, the first conveyor member **44** comprises a series of idle rollers **55** arranged at one end of respective arms **56** hinged onto support bars **57** to form (gravity) pressing means for the paper sheets **51**.

[0079] Instead, the second conveyor member **45** comprises a belt presser **58** placed above the conveyor belts **46** and comprising belts **59** wound about pulleys **60** supported idly on shafts **61**. By virtue of the action of preloaded elastic means (not visible), the belt presser **58** keeps the sheet **51** adhered to the conveyor belts **46**, both in stationary and dynamic phases of the conveyor member **45**.

[0080] The second conveyor member **45** comprises an end proximal **45***a* to the first conveyor member **44** and a distal end **45***b*. Associated with the distal end **45***b* is a guide member **62** of the sheet **51** arranged in a substantially vertical position and comprising a shaft **63** supporting idle pulleys **64**. Drive belts **65** are wound on the pulleys **64** and on corresponding pulleys supported idly on the shaft **61** arranged at the distal end **45***b* of the second conveyor member **45**.

[0081] The belts **36**, **46**, **59**, **65** intended to come in contact with sheet **51** are made of a material with a high friction index, e.g., natural rubber (thickness 1 mm).

[0082] The guide member **62** ends above the contact point between the second drive roller **9** of the winding machine and the winding mandrel **11** of the roll **50**.

[0083] The introduction device **3** can be supported on appropriate actuators (e.g., hydraulic jacks) which allow it to be moved between a raised maintenance position and a lowered operating position (shown in the figures).

[0084] The operation of the introduction device **3** of the paper sheet **51** into the roll **50** being wound will now be described.

[0085] The paper sheets **51**, taken from the feeding tray **31**, are loaded sequentially by the feeder **32** onto the feeder member **33**, so that a sheet **51** comes to rest on the second conveyor member **45**.

At the same time, the continuous strip material S is wound onto the respective winding mandrel **11** which is located at the second drive roller **9** of the winding machine. When a first set of turns **53** has been wound (see FIGS. **8-9**), the second conveyor member **45** is activated, so that sheet **51** is inserted between second drive roller **9** and winding mandrel **11** in a homokinetic condition with the winding speed of the continuous strip material S.

[0086] At this point, a new sheet **51** will be loaded onto the second conveyor member to introduce it into the next roll **50**, which will be wound on a different winding mandrel **11** in the manner described above.

[0087] The sheet **51**, which, as mentioned, can be A4 or similar in size and thus generally have a rectangular shape, is placed with the long side transverse to the winding direction. In any case, regardless of its shape, the size thereof will preferably be such as to come out of the roll **50**, as shown in FIGS. **8** and **9**.

[0088] Therefore, the invention further relates to a method of winding a coreless roll **50** of a continuous strip material S comprising the steps of: [0089] providing a winding machine for coreless rolls of the continuous strip material S, the winding machine being preferably that described above; [0090] winding on a mandrel of the winding machine a first set of turns **53** of the continuous strip material S; [0091] introducing a reinforcing paper sheet **51** into the roll **50** being wound while winding a second set of turns **54**; [0092] singularizing the roll **50** thus formed.

[0093] Preferably, the paper sheet **51** is introduced so that portions of the sheet **51** protrude from the axial ends **50***a* of the roll **50**.

[0094] Preferably, the paper sheet **51** comprises information such as an advertising inscription, a cooking recipe, or other, on at least one surface thereof.

[0095] Although the paper sheet **51** has been described so far, it is apparent that the sheet **51** could also be made of a different material, such as a plastic material of appropriate thickness to achieve a stiffening of the roll **50**, for example.

[0096] The advantages of the present invention are apparent from the above description.

[0097] The coreless roll **50** of continuous strip material S, e.g., food-grade aluminum or alloys thereof or food-grade paper, is reinforced by the paper sheet **51**, without requiring a cardboard core of excessive cost, weight or size.

[0098] The introduction device **3** coupled to the coreless winding machine makes such an operation quick and easy, increasing plant productivity.

[0099] It is apparent that only some particular embodiments of the present invention have been described, to which those skilled in the art will be able to make all those changes required to adapt it to particular applications, without however departing from the scope of protection of the present invention.

Claims

1. A device for introducing a reinforcing paper sheet into a coreless roll being wound with a continuous strip material by a coreless winding machine, comprising: a feeding tray for paper sheets; a feeder of the paper sheets taken from the feeding tray; an introduction member of a paper sheet loaded into a roll being wound in said winding machine.

2. The device according to claim 1, wherein the introduction member of the sheet comprises a first conveyor member and a second conveyor member placed in sequence along an inclined path, wherein the first conveyor member is inclined with a gradient less than a gradient of the second conveyor member.

3. The device according to claim 2, wherein both the first and second conveyor members comprise conveyor belts wound about pulleys supported on shafts rotating by independent drives, wherein the drive of the second conveyor member is configured to operate at a speed to move the sheet homokinetically with respect to a winding speed of the roll.

4. The device according to claim 2, wherein the first conveyor member comprises a series of idle rollers arranged at one end of respective arms hinged onto support bars to form gravity pressing means for the paper sheets.
5. The device according to claim 2, wherein the second conveyor member comprises a belt presser placed above the conveyor belts and comprising belts wound about pulleys supported idly on shafts.
6. The device according to claim 2, wherein the second conveyor member comprises a proximal end with respect to the first conveyor member and a distal end, wherein said distal end is associated with a guide member for the sheet arranged in a substantially vertical position and comprising a shaft which supports idle pulleys, a drive belt being wound on said pulleys and on corresponding pulleys supported idly on a shaft arranged at the distal end of the second conveyor member.
7. The device according to claim 1, wherein the feeding tray comprises containment edges and an open bottom placed above a proximal end of the feeder, wherein the containment edge placed straddling the feeder ends proximate a gripping surface of the feeder without coming into contact with the feeder, to allow the paper sheet to be singularized and taken from a stack of sheets from below.
8. The device according to claim 1, wherein the feeder comprises an ascending belt system running about motorized pulleys and idle pulleys, wherein the belt system changes the gradient thereof proximate a distal end of the feeder, forming a substantially horizontal segment.
9. The device according to claim 8, wherein a plurality of idle rollers is placed above the belt system, at one end of respective arms hinged onto support bars to form gravity pressing means for the paper sheets transiting on the feeder.
10. A coreless-type winding machine comprising a device according to claim 1, wherein said winding machine comprises: a coil-holder roller which rotationally supports a coil of continuous strip material; a first motorized drive roller and a second motorized drive roller, wherein the second drive roller is coupled to a pressing roller so that the continuous strip material passes between said two rollers; a plurality of motorized winding mandrels, placed on a revolver member, said revolver member being discontinuously rotatable, to present a different winding mandrel from time to time at the second drive roller; wherein the guide member of the second conveyor member of the device ends above a contact point between the second drive roller and the winding mandrel of the roll being wound; and wherein said winding mandrels have, along a cylindrical surface thereof, fixed circular arc sectors alternating with retractable circular arc sectors, which form a cylindrical surface during the step of winding to form the roll and retract radially during the step of extracting the roll.
11. A method of winding a coreless roll of a continuous strip material comprising the steps of: providing a winding machine for coreless rolls of said continuous strip material; winding on a mandrel of said winding machine a first set of turns of said continuous strip material; introducing a reinforcing paper sheet into said winding roll while winding a second set of turns; singularizing a roll formed.
12. The method according to claim 10, comprising the step of providing a coreless winding machine comprising a device for introducing a paper sheet into a coreless roll being wound with a continuous strip material by a coreless winding machine, the device comprising: a feeding tray for paper sheets, a feeder of the paper sheets taken from the feeding tray, an introduction member of a paper sheet loaded into a roll being wound in said winding machine.
13. A coreless roll of a continuous strip material, comprising a central hole surrounded by a first set of turns of said continuous strip material and a second set of turns of said continuous strip material, wherein the first set of turns comprises a smaller number of turns than the second set of turns, and wherein a paper sheet is placed between the first set of turns and the second set of turns.
14. The roll according to claim 13, wherein the wound paper sheet protrudes from two axial ends of the roll.
15. The roll according to claim 13, wherein the paper sheet comprises information, including

advertising lettering, recipes, or contest indications.
16. The device according to claim 1, wherein the introduction member of the sheet comprises a first conveyor member and a second conveyor member placed in sequence along an inclined path, wherein the first conveyor member is inclined with a gradient less than the second conveyor member, and wherein an angle formed by the plane of the second conveyor member with respect to a horizontal plane is greater than 45°.
17. The device according to claim 1, wherein the introduction member of the sheet comprises a first conveyor member and a second conveyor member placed in sequence along an inclined path, wherein the first conveyor member is inclined with a gradient less than the second conveyor member, and wherein an angle formed by the plane of the second conveyor member with respect to a horizontal plane is greater than or equal to 70°.
18. A coreless roll of a continuous strip material, comprising food grade aluminum or alloys thereof or food grade paper, comprising a central hole surrounded by a first set of turns of said continuous strip material and a second set of turns of said continuous strip material, wherein the first set of turns comprises a smaller number of turns than the second set of turns and comprises between 1 and 9 turns, and wherein a paper sheet is placed between the first set of turns and the second set of turns.
19. The roll according to claim 13, wherein the wound paper sheet protrudes from two axial ends of the roll by 2-7 mm.